



Digital IC Design

Linux and EDA Tool

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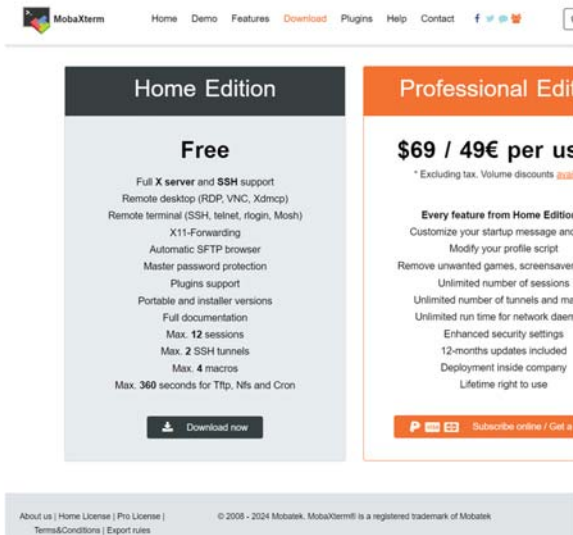


Outline

- ☐ Connection
- ☐ Linux Command
- ☐ RTL Simulation
- ☐ Notice

Connection

- ❑ 1. SSH
 - MobaXterm



The image shows the MobaXterm website with two main sections: Home Edition and Professional Edition.

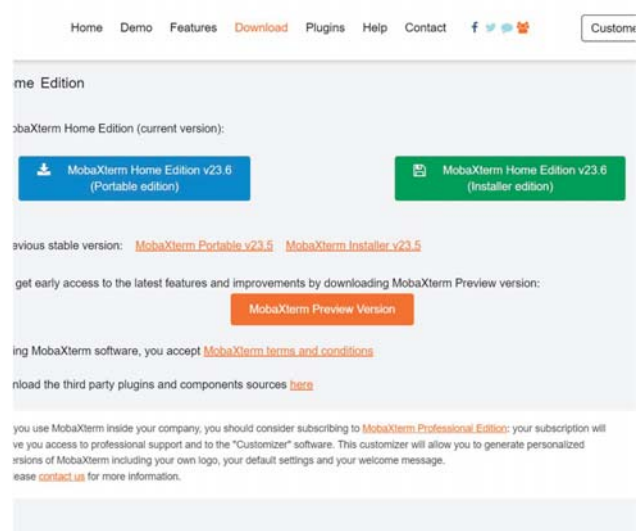
Home Edition (Free):

- Full X server and SSH support
- Remote desktop (RDP, VNC, Xdmcp)
- Remote terminal (SSH, telnet, rlogin, Mosh)
- X11-Forwarding
- Automatic SFTP browser
- Master password protection
- Plugins support
- Portable and installer versions
- Full documentation
- Max. 12 sessions
- Max. 2 SSH tunnels
- Max. 4 macros
- Max. 360 seconds for Tftp, Nfs and Cron

Professional Edition (\$69 / 49€ per user):

- Every feature from Home Edition
- Customize your startup message and Modify your profile script
- Remove unwanted games, screensaver
- Unlimited number of sessions
- Unlimited number of tunnels and max
- Unlimited run time for network daemon
- Enhanced security settings
- 12-months updates included
- Deployment inside company
- Lifetime right to use

Buttons: Download now, Subscribe online / Get a trial



The image shows the MobaXterm website's download and features section.

Download:

- MobaXterm Home Edition v23.6 (Portable edition)
- MobaXterm Home Edition v23.6 (Installer edition)

Previous stable version: [MobaXterm Portable v23.5](#) [MobaXterm Installer v23.5](#)

Get early access to the latest features and improvements by downloading MobaXterm Preview version:

[MobaXterm Preview Version](#)

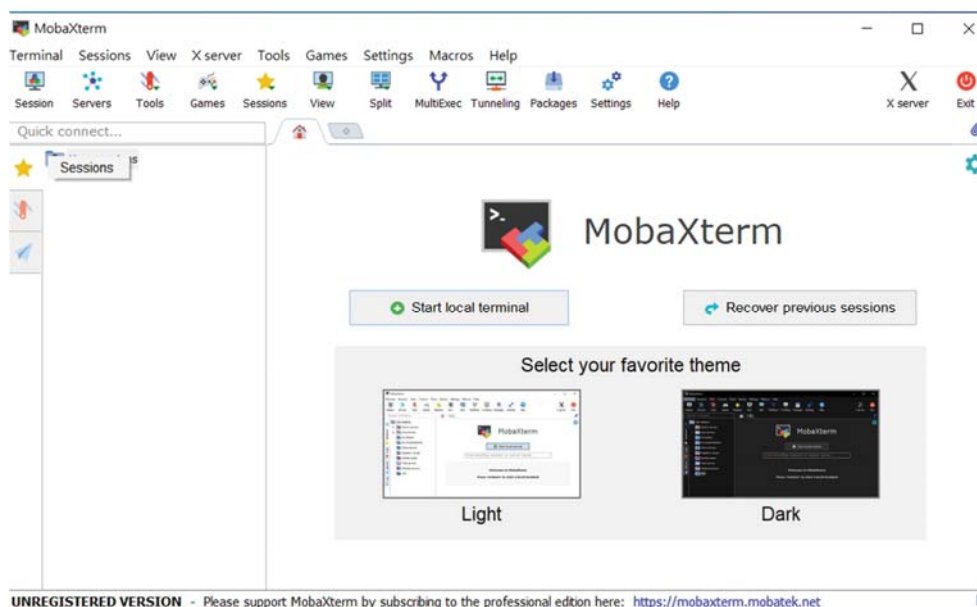
Using MobaXterm software, you accept [MobaXterm terms and conditions](#)

Download the third party plugins and components sources [here](#)

If you use MobaXterm inside your company, you should consider subscribing to MobaXterm Professional Edition: your subscription will give you access to professional support and to the "Customizer" software. This customizer will allow you to generate personalized versions of MobaXterm including your own logo, your default settings and your welcome message. Please [contact us](#) for more information.

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SSH



The image shows the MobaXterm software interface. The top menu bar includes: Terminal, Sessions, View, X server, Tools, Games, Settings, Macros, Help. The top toolbar includes: Session, Servers, Tools, Games, Sessions, View, Split, MultiExec, Tunneling, Packages, Settings, Help, X server, Exit.

Quick connect...

Sessions

MobaXterm

Start local terminal **Recover previous sessions**

Select your favorite theme

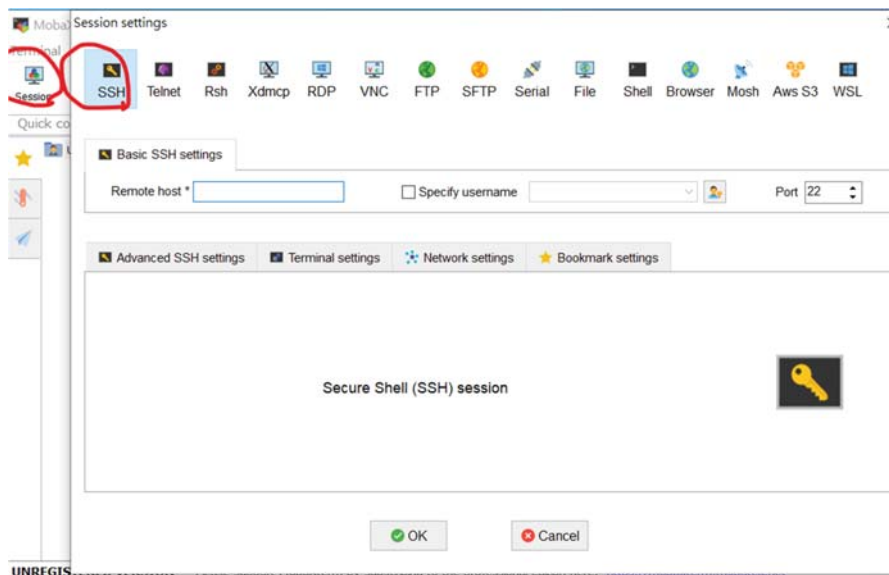
Light **Dark**

UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>

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SSH (cont'd)

- ☐ Remote host :163.25.98.12
- ☐ Tick Specify username : your account
- ☐ Port : 50022



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SSH (cont'd)

```
? MobaXterm Personal Edition v23.5 ?
(SSH client, X server and network tools)

> SSH session to acc00@163.25.98.12
? Direct SSH      : ✓
? SSH compression : ✓
? SSH-browser     : ✓
? X11-forwarding  : ✗ (disabled or not supported by server)

> For more info, ctrl+click on help or visit our website.

You are required to change your password immediately (root enforced)
Last login: Fri Mar 1 09:39:11 2024 from 192.168.71.242
WARNING: Your password has expired.
You must change your password now and login again!
Changing password for user acc00.
Changing password for acc00.
(current) UNIX password:
New password:
BAD PASSWORD: The password is shorter than 8 characters
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
```

```
? MobaXterm Personal Edition v23.5 ?
(SSH client, X server and network tools)

> SSH session to acc00@163.25.98.12
? Direct SSH      : ✓
? SSH compression : ✓
? SSH-browser     : ✓
? X11-forwarding  : ✓ (remote display is forwarded through SSH)

> For more info, ctrl+click on help or visit our website.

Last login: Fri Mar 1 09:39:40 2024 from 192.168.71.242
/usr/bin/xauth: file /home/DEIC/acc00/.Xauthority does not exist
Welcome to elecgamma.cgu.edu.tw
Kernel version 3.10.0-1062.el7.x86_64 of CentOS Linux release 7.7.1908 (Core)

Setting Up the EDA Tools Environment ... Completed

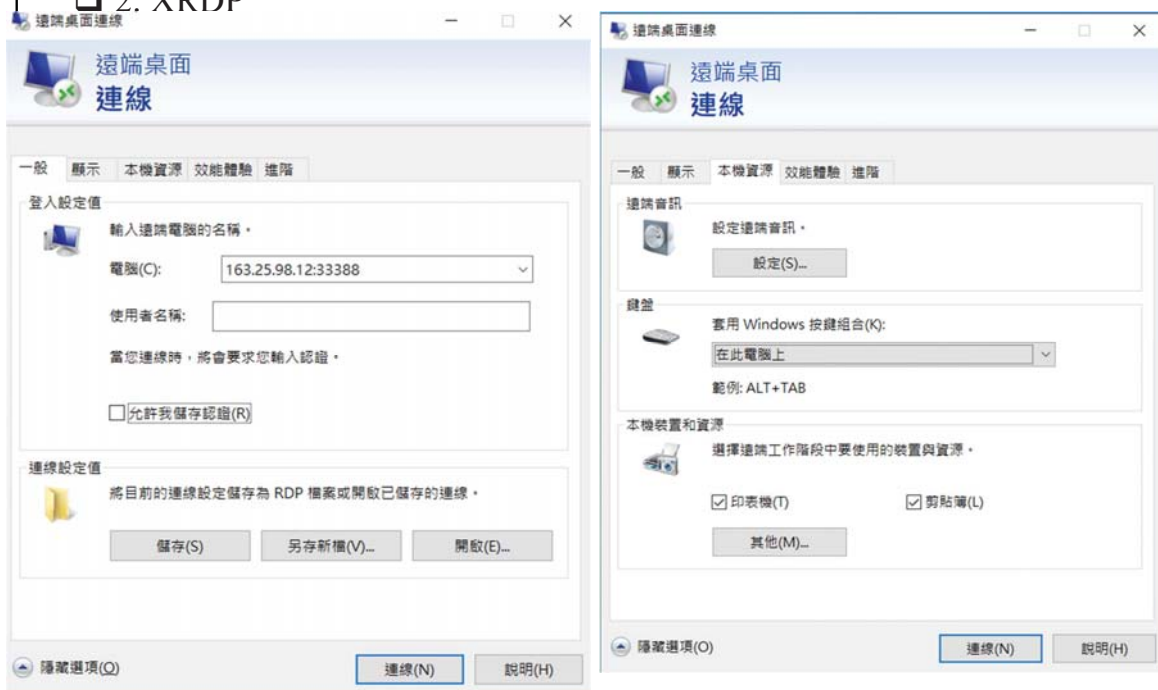
#####
## IC6 is the default version                                ##
## If you want to run IC5,                                    ##
## please set the IC_MAJOR_VERSION to 5 in ~/.cshrc by yourself. ##
##                                                            ##
## Process Design Kit Path: /cad/PDK                         ##
## Cell-based Design Kit Path: /cad/CBDK                     ##
#####

[acc00@elecgamma ~]$
```

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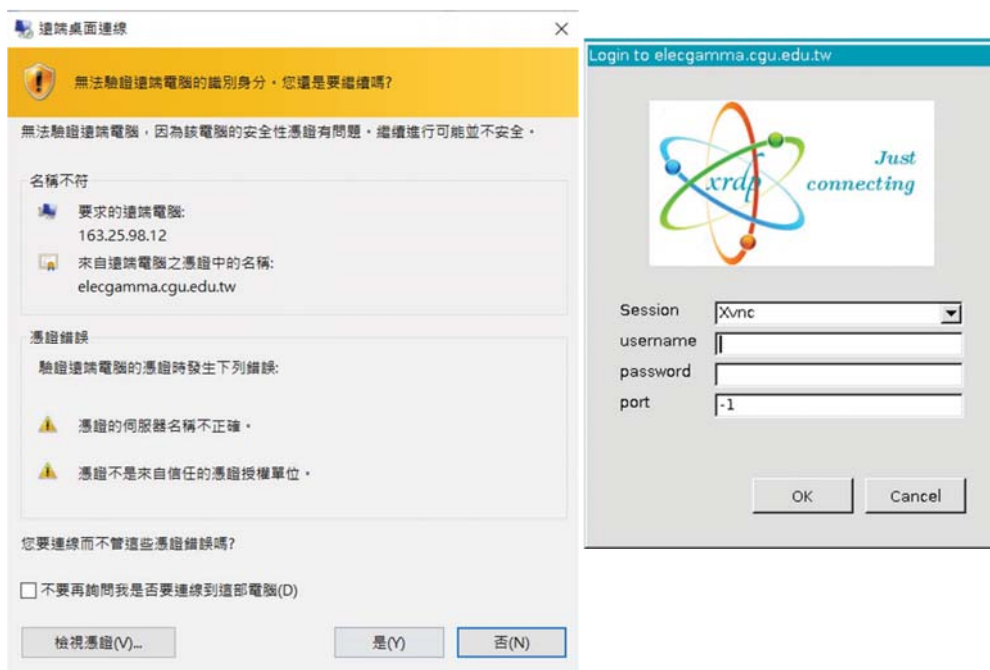
Connection

2. XRDP



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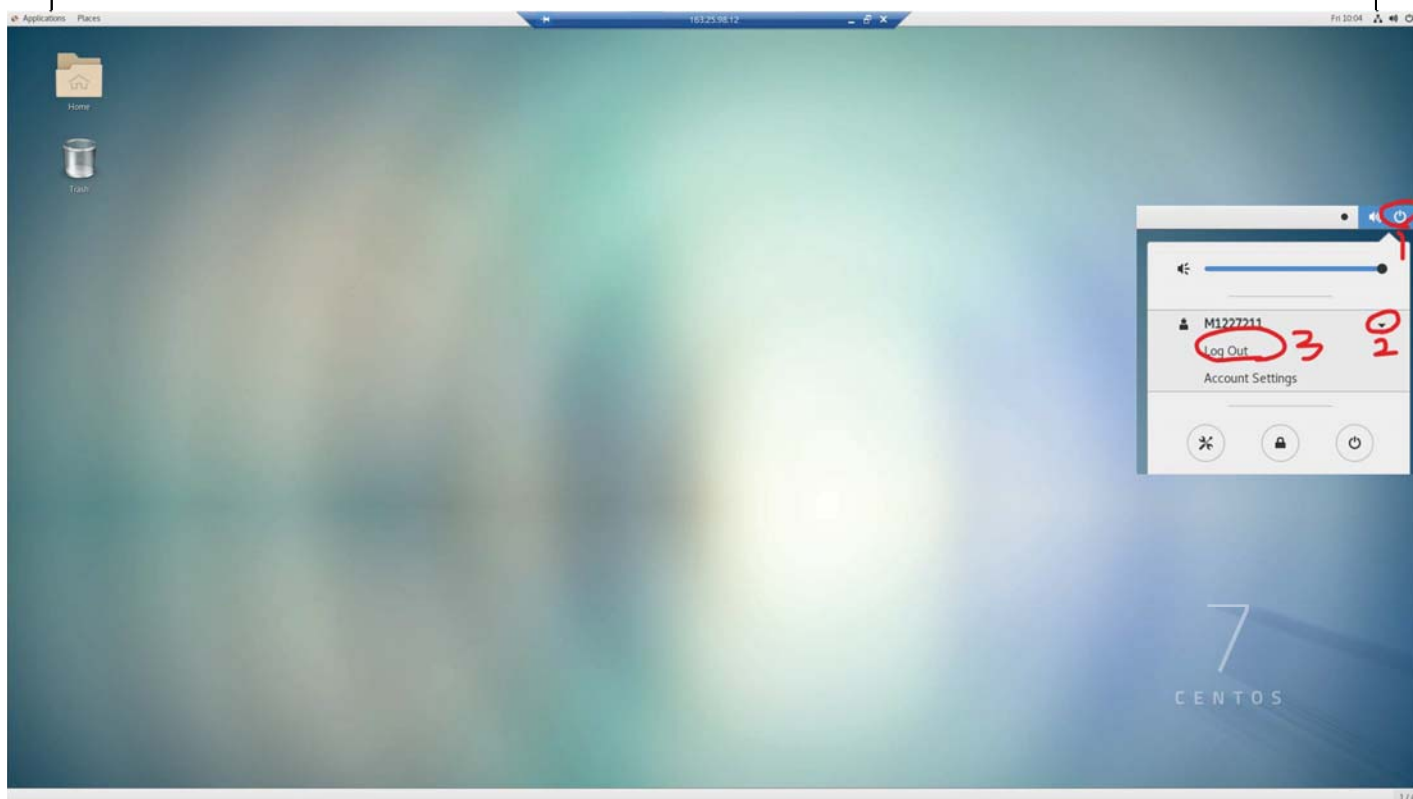
XRDP



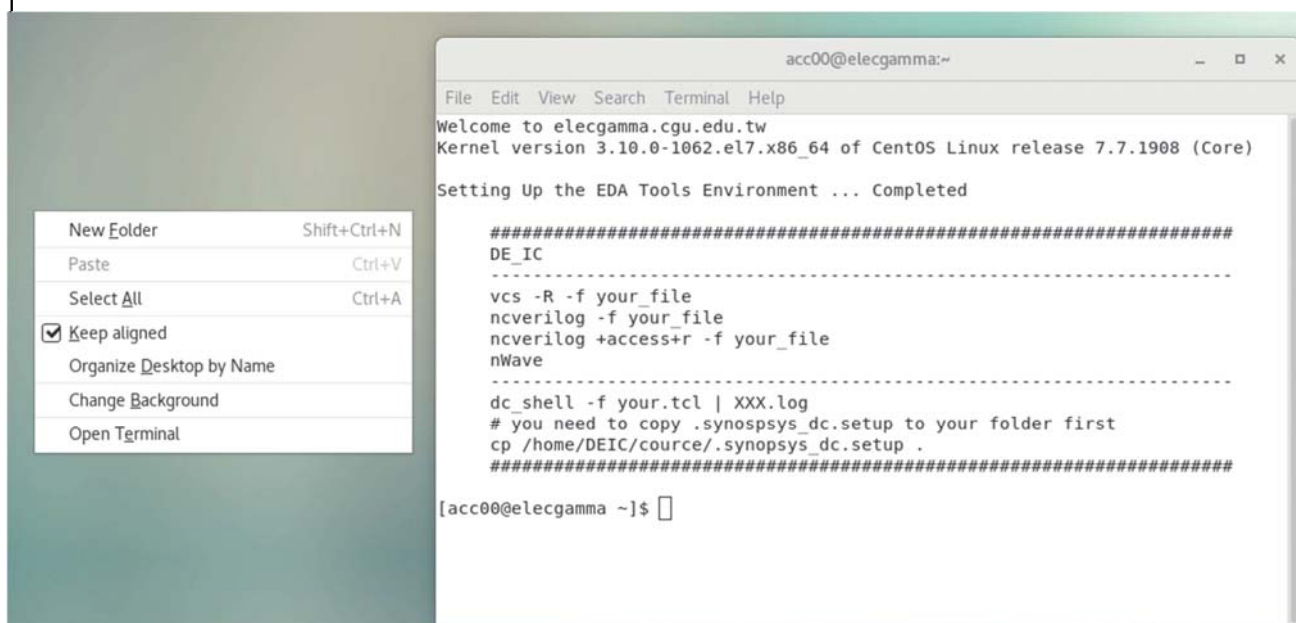
8



XRDP (cont'd)

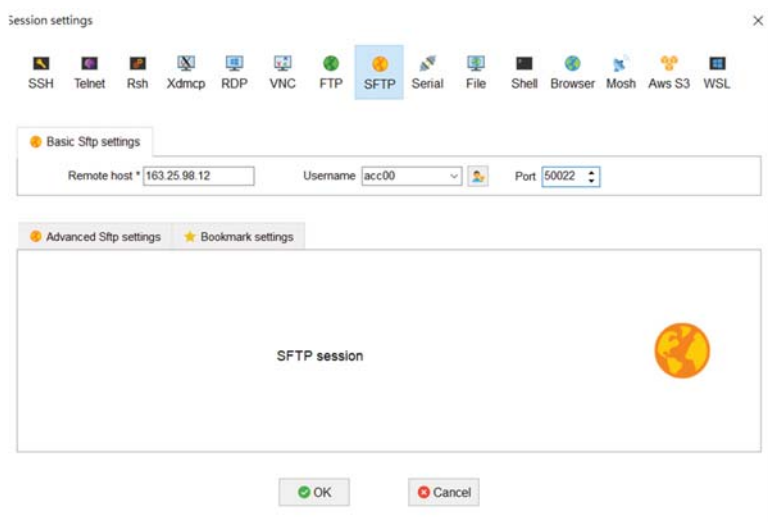


XRDP (cont'd)



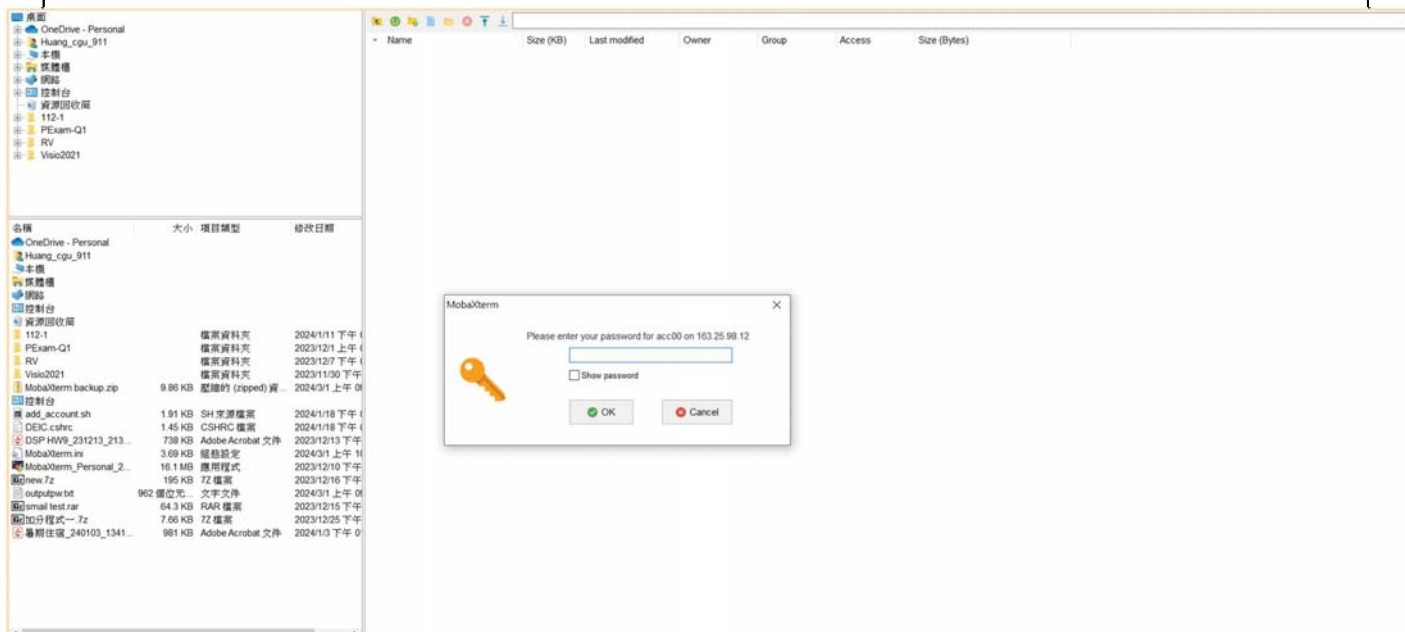
SFTP

- ☐ Remote host :163.25.98.12
- ☐ username : your account
- ☐ Port : 50022



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SFTP (cont'd)



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- 到Homework 的資料夾
- 回到使用者根目錄
- 到Homework/test_lab0/
- 回到上一層資料夾

查看目前絕對路徑

Homework4 Homework5

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Linux Command (cont'd)

- ❑ mkdir 創建資料夾
- ❑ rmdir 移除資料夾(若資料夾內有檔案無法使用)
- ❑ **rm -rf** 移除資料夾(資料夾內有檔案可使用) 慎用
- ❑ cp copy
 - cp 來源檔 目的地(或名稱)
 - cp -r 複製資料夾
 - [acc00@elecgamma test]\$ cp ../test_lab0/READ_ME_simulation.txt .
 - [acc00@elecgamma test]\$ ls
READ_ME_simulation.txt
 - [acc00@elecgamma test]\$ cp ../test_lab0/READ_ME_simulation.txt Hello.txt
 - 複製並將名稱改為Hello.txt
 - [acc00@elecgamma test]\$ ls
Hello.txt READ_ME_simulation.txt
- ❑ mv 搬運資料夾或檔案
 - mv 來源檔 目的地(或名稱)
 - mv intro manual/chap1 將檔案intro搬運至manual資料夾中並改名chap1

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Linux Command (cont'd)

- ❑ gedit 文字編輯器
 - gedit abc.xxx 使用gedit開啟 abc.xxx，若檔案不存在，建立abc.xxx
- ❑ vi 文字編輯器
 - vi abc.xxx 使用vi開啟 abc.xxx，若檔案不存在，建立abc.xxx
 - vi command

i	進入編輯模式
Esc	退出編輯模式
:q	不進行儲存並離開vi
:wq	儲存並離開vi

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RTL Simulation

- ☐ vcs
 - vcs -R -f sim.f
- ☐ ncverilog
 - ncverilog -f sim.f
 - ncverilog +access+r -f sim.f



- ☐ nWave
 - nWave

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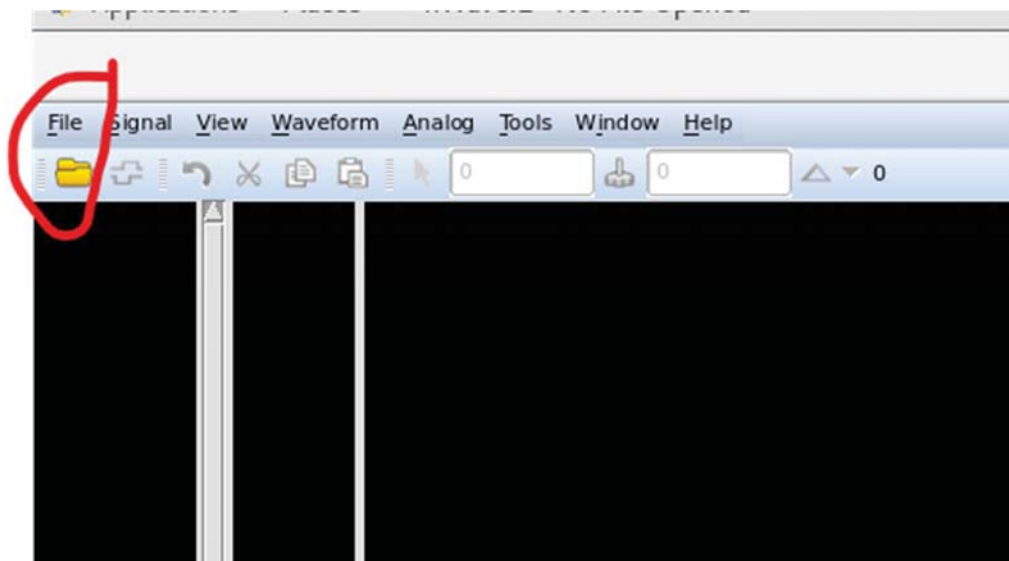


RTL Simulation (cont'd)

- | | |
|---|--|
| <ul style="list-style-type: none">☐ vcs <pre>initial begin \$fsdbDumpfile("test.fsdb"); \$fsdbDumpvars(0, test_rtb); end</pre> | <ul style="list-style-type: none">☐ ncverilog <pre>initial begin \$dumpfile("test.vcd"); \$dumpvars(0, test_rtb); end</pre> |
|---|--|

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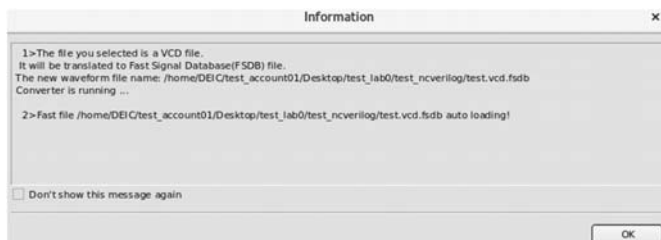
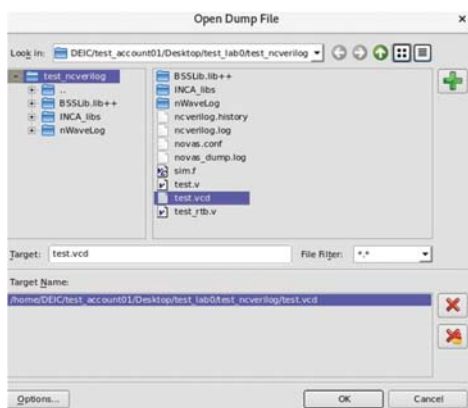
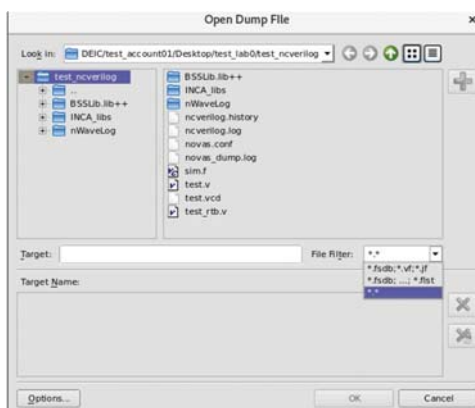
nWave



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nWave (cont'd)

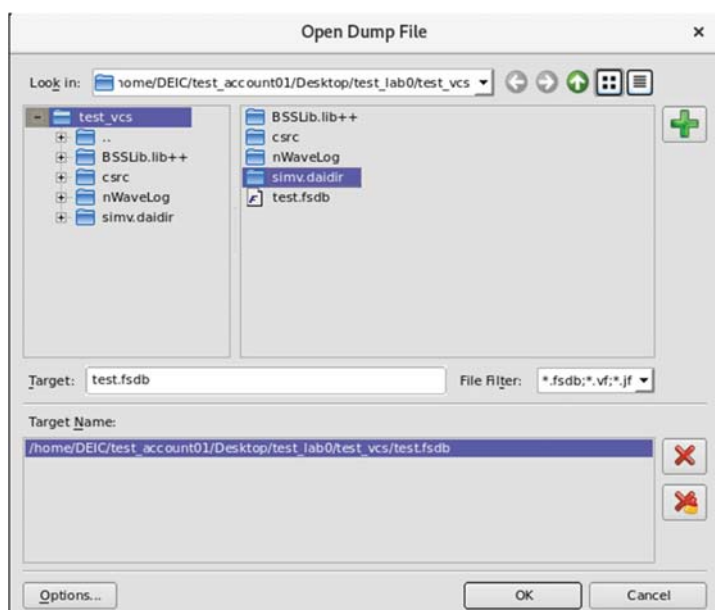
☐ ncverilog(.vcd檔)



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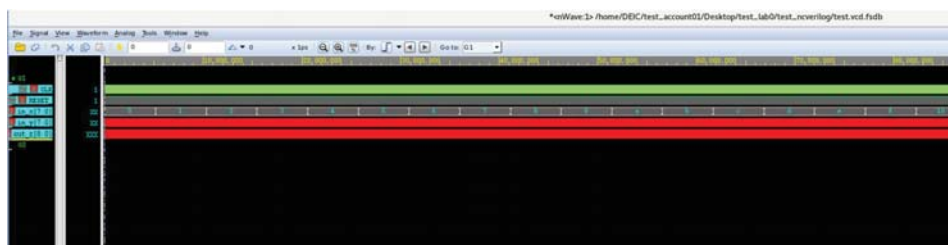
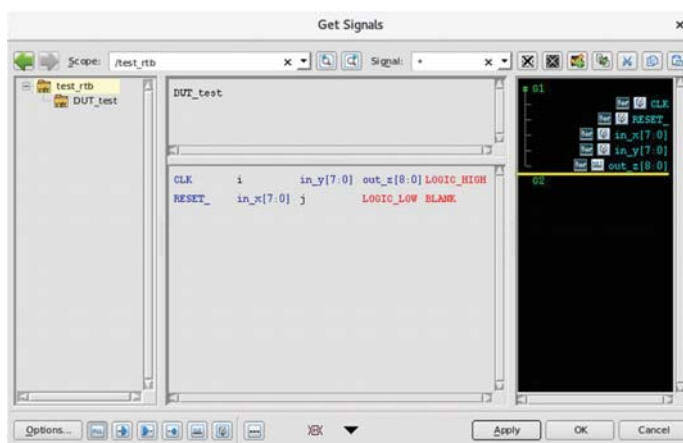
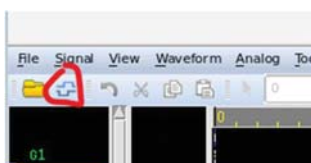
nWave (cont'd)

❑ vcs(.fsdb檔)



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nWave (cont'd)



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Notice

❑ HW作業繳交項目

- 書面報告

❑ E-learning繳交項目

- 請新增一資料夾存放下列項目(1. 與 2.)，並壓縮(.zip or .rar 等)再進行繳交
- 1. 書面報告電子檔
- 2. Verilog RTL simulation
 - All files of xxx_rtl.v and xxx_rtb.v
 - sim.f

❑ For example:

- ~HW1/Report

```
/RTL simulation  /sim.f
                  /xxx_rtb.v
                  /xxx_rtl.v
```

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Notice (cont'd)

❑ SSH 無法連線

- 宿舍區請提供學校所分配IP (172.2X.XXX.XXX)
 - 查詢方法: cmd (命令提示字元) --- ipconfig

```
乙太網路卡 乙太網路:
  連線特定 DNS 尾碼 . . . . . : 
  IPv4 位址 . . . . . : 192.168. .
  子網路遮罩 . . . . . : 255.255.255.0
  預設閘道 . . . . . : 192.168. .

無線區域網路介面卡 區域連線* 1:
  媒體狀態 . . . . . : 媒體已中斷連線
  連線特定 DNS 尾碼 . . . . . : 

無線區域網路介面卡 區域連線* 10:
  媒體狀態 . . . . . : 媒體已中斷連線
  連線特定 DNS 尾碼 . . . . . : 

無線區域網路介面卡 Wi-Fi:
  媒體狀態 . . . . . : 媒體已中斷連線
  連線特定 DNS 尾碼 . . . . . : 

乙太網路卡 藍牙網路連線:
  媒體狀態 . . . . . : 媒體已中斷連線
  連線特定 DNS 尾碼 . . . . . :
```

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Notice (cont'd)

- ❑ SSH 無法連線(實驗室)
 - 實驗室請提供實驗室實體IP (163.25.XXX.XXX)
 - 或提供由系所分配之虛擬IP (ex.192.168.71.XXX)
- ❑ 請勿提供由各實驗室或自行購買路由器所分配之IP (DHCP)
 - 華碩 192.168.50.XXX
 - tplink、Dlink 192.168.0.XXX
 - Other 192.168.1.XXX, 192.168.10.XXX
- ❑ 路由器請登入進行查詢



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Notice (cont'd)

- ❑ 密碼遺失或XRDP當機
 - 請用**學校email**傳訊息至以下email
 - M1327224@cgu.edu.tw
 - M1327212@cgu.edu.tw
 - M1327213@cgu.edu.tw
 - **不接受其餘email，使用其餘email將會忽視**
 - 密碼遺失請在主旨處打上“DICD 學號:XXX 帳號XXX 密碼遺失”
 - 密碼會利用email回傳一組臨時密碼，請登入後自行修改
 - 更改指令: passwd
 - XRDP當機請先在 SSH輸入指令，再上線重試
 - kill all -u username(自己帳號名稱)
 - 如果XRDP還是當機請在主旨處打上“DICD 學號:XXX 帳號XXX XRDP當機”
 - 處理完成後會回傳一封email，請在上線重試

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Notice (cont'd)

❑ Password Lost or XRDP Crash

- Please send a message **using your school email** to the following addresses only:
 - M1327224@cgu.edu.tw
 - M1327212@cgu.edu.tw
 - M1327213@cgu.edu.tw
- **Other email addresses will not be accepted. Messages sent from non-school emails will be ignored.**
- Password Lost
 - Email Subject Format: **“DICD StudentID: XXX Account: XXX Password Lost”**
 - A temporary password will be sent back via email.
 - Please log in and change it immediately
 - Use the following command to change your password: passwd
- If XRDP crashes:
 - First, log in via SSH and enter: **kill all -u username**
- If XRDP is still not working:
 - Email Subject Format: **“DICD StudentID: XXX Account: XXX XRDP Crash”**
 - After the issue is fixed: You will receive a confirmation email. Please try logging in again.

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Lab

- ❑ cp -R ~/. ./course/test_lab0 ●

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